



Goal: $\mathbb{E}[Y] = \underset{a \in \mathbb{R}}{\operatorname{argmin}} \mathbb{E}[(Y-a)^2]$

$$\mathbb{E}[(Y-a)^2] = \mathbb{E}[(Y - \mathbb{E}[Y] + \mathbb{E}[Y] - a)^2]$$

$$= \underbrace{\mathbb{E}[(Y - \mathbb{E}[Y])^2]}_{\operatorname{Var}[Y]} + \underbrace{\mathbb{E}[2(Y - \mathbb{E}[Y])(\mathbb{E}[Y] - a)]}_0 + \mathbb{E}[(\mathbb{E}[Y] - a)^2]$$

$$\min_{a \in \mathbb{R}} \mathbb{E}[(Y-a)^2] = \operatorname{Var}[Y] + \min_{a \in \mathbb{R}} \mathbb{E}[(\mathbb{E}[Y] - a)^2]$$

$$\min_{g \in \mathcal{G}} \mathbb{E}[(Y - g(X))^2]$$

$$\mathbb{E}[(Y - g(X))^2] = \mathbb{E}[(Y - \mathbb{E}[Y|X] + \mathbb{E}[Y|X] - g(X))^2]$$

$$= \mathbb{E}[(Y - \mathbb{E}[Y|X])^2] + 2 \mathbb{E}[(Y - \mathbb{E}[Y|X])(\mathbb{E}[Y|X] - g(X))]$$

$$+ \mathbb{E}[(\mathbb{E}[Y|X] - g(X))^2]$$

$$\mathbb{E}[\mathbb{E}[(Y - \mathbb{E}[Y|X])(\mathbb{E}[Y|X] - g(X)) | X]] = 0$$

$$\Rightarrow \boxed{g^*(X) = \mathbb{E}[Y|X]}$$